

Q and P: Institutions, Agents, and Stopping Rules

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Adversarial Review: Structured Disagreement for Grounded Agentic Code Review [arXiv:2608.18167](#)

Research phase: round 4, verification pending, 3 ITERATE, 0 REVISE

The pair in two sentences

Q studies LLM traders in a continuous double auction. It reports slower or absent convergence, and lower allocative efficiency, than in the human benchmark. P studies Adversarial Review (AR), in which a reviewer and a critic exchange structured judgments before a coding agent revises its work, and reports gains alongside a false-consensus failure mode.

The connexion pursued

The question was whether observed multi-agent performance comes from what the agents know or from the institution, especially its stopping rule and its admissible messages. A zero-intelligence control can separate the two. Q permits a constrained random-trader benchmark. P suggests analogous controls: a critic whose verdicts are random, or a reviewer that is allowed to iterate alone [\[ledger 11, 14, 16, 18\]](#).

After the verifier’s second ITERATE [\[ledger 33\]](#), the owner authorised one more iteration on the matched P ablation. It was not to use paid model campaigns, and it was not to claim the causal gap closed without evidence [\[ledger 40\]](#). This round therefore stated the P question as an identity over paired arms run on the same tasks:

- Zero-shot (ZS);
- Single-reviewer (SR), which edits once;
- an R-only loop that iterates until the reviewer approves at first pass (RO);
- AR;
- AR with a content-free critic whose DISAGREE rate matches the realised AR rate (ARZ).

Then

$$AR - SR = (RO - SR) + (ARZ - RO) + (AR - ARZ).$$

The first term measures re-review and stopping. The second measures the extra edits that a critic forces whatever its content. The third is the value of the critic’s information [\[ledger 43, 49, 52\]](#). This decomposition is the P counterpart of reading Q’s LLM markets against the ZI-C floor, with omission and commission scored separately [\[ledger 52, 54\]](#).

What was found

Q: a zero-intelligence floor (earlier rounds, still standing)

The firm result concerns Q alone. Two independent implementations of Q’s stated mechanism give constrained zero-intelligence traders (ZI-C) an efficiency near 0.96, about 5.8–5.9 trades per round and a mean transaction price of 2.00. Every reported LLM cell in Q has efficiency 0.36–0.91 [\[ledger 3, 5\]](#). The comparison is robust to the simulated quote range: efficiency stays at 0.943–0.957 [\[ledger 5\]](#). On a finer horizon grid, ZI-C reaches the five-round LLM mean efficiency after only about 53 draws for GPT Large, 80 for Gemini Large and 131 for GPT Small, whereas Q allows 300 draws [\[ledger 8\]](#).

Claim 1. *In Q 's mechanism, and conditional on the LLMs complying with the reservation bound stated in their prompt, budget-constrained random traders extract about 0.96 of the maximum surplus, more than any reported LLM market [ledger 3, 5, 18, 31].*

The surplus accounting explains the benchmark. The maximum feasible surplus is 7.50. Define

$$1 - E = O + C,$$

where O is the surplus forgone because intramarginal traders were left out, divided by 7.50, and C is the surplus destroyed by extramarginal participation, divided by 7.50. The five efficient pairs contribute 2.50, 2.00, 1.50, 1.00, 0.50. So a round with $k = 1, \dots, 5$ trades has efficiency at most 0.33, 0.60, 0.80, 0.93, 1.00 respectively [ledger 5, 7]. ZI-C has $O = 0.009$ and $C = 0.032$. A stylised one-cent “creeper” has an omission of about 0.82. Making acceptance the default raises volume but produces a commission of 0.10–0.17 and an efficiency of only 0.71–0.79 [ledger 4]; an independent implementation gives 0.79–0.82 with a commission of 0.10–0.11 [ledger 12]. Thus a cheap terminal move can exchange missed trades for bad trades without recovering the ZI-C result. The direction of failure depends on the model. The reported GPT Large and Gemini Large markets mainly omit trades, but GPT Small and the unreported Gemini runs also show commission [ledger 6, 7].

P: the confound (earlier rounds, still standing)

None of this establishes the proposed connexion to P. In P, Single-reviewer and Two-reviewers edit once, while MARS and AR re-review revised artifacts. MARS and AR are also the only methods above the 75–77% cluster [ledger 11, 16, 22]. Consequently P cannot separate the informational value of reviewer–critic interaction from re-review, stopping and adaptive compute. A four-parameter toy model shows only that the confound is plausible. When one review–edit cycle is calibrated to have zero net effect, iterating until approval adds 0.3–9.1 percentage points (median 2.6). A random critic that blocks approval with probability 0.3 lowers pass rates in every parameter set. The toy is not evidence about P's tasks [ledger 12, 36].

This round: access to P

P is on arXiv as v1 only (16 August 2026), with no code or data link. Its Appendix A says that the prompts “will be released alongside the implementation upon acceptance”. Searches for an author repository found only unrelated projects named adversarial-review [ledger 43, 44]. P does not list its 105 LiveCodeBench (LCB) stdin tasks, its per-task outcomes, its LCB outer cap or its critic disagreement rates. LCB release v6 has 112 stdin AtCoder tasks (60 hard, 4 January to 6 April 2025), and no contiguous contest-date window of it gives 105 tasks with 57 hard. P's task set therefore remains unidentified [ledger 43, 45]. P's model, Claude Sonnet 4.5, is paid and was excluded [ledger 40, 43]. An exact reproduction, or a matched ablation on P's own tasks, model and harness, is still impossible, and no paid calls were made [ledger 43, 44, 54].

This round: what P's reported aggregates identify

The rounded percentages in P's Table 1 each admit exactly one count out of 105: ZS 81, Self-Refine 81, Single-reviewer 81, Two-reviewers 79, MARS 86 and AR 91. The prose value of 85% for MARS would be 89 [ledger 45, 46]. On the 48 non-hard tasks, AR solves 48 and Single-reviewer 45. So 7 of AR's 10 extra tasks over Single-reviewer are on the hard tier, as P says [ledger 45].

P reports no per-task pairing, so the peers computed exact two-sided McNemar p -values over every joint table compatible with the marginals. The lower ends follow in closed form. Let b and c be the numbers of discordant tasks favouring each method. The marginals fix $\delta = b - c$, and the exact test gives

$$p = \min\left(1, 2 \sum_{i \leq \min(b,c)} \binom{b+c}{i} 2^{-(b+c)}\right),$$

which is smallest at $c = 0$, where $p = 2^{1-\delta}$. The upper ends come from the largest discordance the marginals allow. The resulting ranges [ledger 45, 46] are:

- AR vs MARS ($\delta = 5$): $p \in [0.0625, 0.49]$; on the hard subset ($\delta = 4$), $p \in [0.125, 0.60]$.
- MARS vs Single-reviewer ($\delta = 5$): $p \in [0.0625, 0.54]$.

- AR vs Single-reviewer ($\delta = 10$): $p \in [0.0020, 0.14]$; on the hard subset ($\delta = 7$), $p \in [0.016, 0.31]$.

Claim 2. *In a single run on 105 tasks, P’s orderings “MARS is clearly higher than the cluster” and “AR beats MARS” cannot reach $p < 0.05$ under any pairing. Only AR over the 77% cluster can, and only if AR loses almost no task that Single-reviewer solves [ledger 45, 46].*

This round found a second prose–table mismatch in P. The prose credits MARS with “+3 hard tasks (43/57)” over Single-reviewer, but Table 1 gives MARS 39/57, and 43/57 is AR’s figure. Both passages are in `inputs/P.tex`: the prose at line 171 and Table 1 [ledger 45, 49].

This round: what P’s token medians imply

P’s Fig. 5a gives median LCB tokens per task of 8000 for ZS, 22000 for Single-reviewer, 28000 for Two-reviewers and 30000 for AR, “estimated from per-method call counts” [ledger 53]. Let g , r , e and c be the token costs of a generation, a review, an edit and a critique. From $g = 8000$, $g + r + e = 22000$ and $g + 2r + e = 28000$, we get $r = 6000$ and $e = 8000$. In AR every edit is followed by a new inner round. The cheapest edited AR task therefore makes six calls (generate, review, critique, edit, review, critique) and costs

$$g + 2r + e + 2c = 28000 + 2c,$$

whereas a task accepted at first pass costs $g + r + c$ [ledger 53]. The median of 105 tasks is the 53rd. If $c > 1000$, every edited task costs more than the median, so at least 53 AR tasks shipped their first version v_0 unedited. AR fails 14 tasks, so at most 14 failing v_0 were accepted, and the +10 over ZS and Single-reviewer comes from at most 52 edited tasks [ledger 53].

Take the ZS failure count $F = 24$ as a proxy for the number of failing AR v_0 . Suppose a stop rule that ignores correctness accepts a tasks. The number of failing tasks it accepts is then hypergeometric, and $P(\text{at most 14 failures accepted}) \geq 0.05$ only for $a \leq 77$. The threshold is $a \leq 89$ for $F = 20$ and $a \leq 67$ for $F = 28$. For example, with $F = 24$ the probability is 0.055 at $a = 77$ and 0.016 at $a = 81$ (`emmy/p_token_bound_out.txt`) [ledger 53].

Claim 3. *If P’s logs show more than about 77 first-pass accepts, then P’s stop decision (R, or R with C) must carry correctness information, whatever the edits do. If $53 \leq a \leq 77$, P’s numbers are compatible with a content-free stop rule combined with effective edits. P does not report a , so this part of the verifier’s question reduces to one number in P’s logs, ideally split by whether v_0 passes [ledger 53, 54].*

This bound rests on two assumptions: that P’s medians are call-count estimates, as its caption says, and that AR’s v_0 has the ZS failure distribution (same model, same generation prompt, single run) [ledger 53].

This round: what a content-free critic can and cannot do

Under P’s first-pass termination, AR stops the outer loop only when R approves and C agrees in the first inner round [ledger 44]. Write $\text{TPR} = P(\text{R approves} \mid \text{correct})$ and $\text{FPR} = P(\text{R approves} \mid \text{buggy})$. A critic that disagrees with probability d , independently of correctness, gives

$$\text{TPR}_{\text{ZI}} = (1 - d) \text{TPR}, \quad \text{FPR}_{\text{ZI}} = (1 - d) \text{FPR}.$$

Let π be the share of correct artifacts under review. The buggy share among shipped artifacts is $(1 - \pi)\text{FPR} / (\pi \text{TPR} + (1 - \pi)\text{FPR})$, because the factor $1 - d$ cancels. The shipped-buggy share is therefore unchanged. Such a critic only sends more artifacts, correct ones included, back for editing [ledger 44, 52]. The real critic adds stopping information if and only if

$$P(\text{C agrees} \mid \text{R approves, correct}) \neq P(\text{C agrees} \mid \text{R approves, buggy}).$$

So AR – ARZ is the sum of stopping information and edit guidance from the critic’s text. The institutional part, the extra forced edits, is exactly what ARZ reproduces [ledger 52]. This argument is analytic and needs no data. It identifies the quantities to measure but does not measure them.

This round: local controlled experiments, with no result yet

Two free experiments were started on a local open-weight model. Emmy’s run (`emmy/ar_ablation.py`) uses:

- gemma4:26b through Ollama, with thinking off and temperature 0.7;
- 40 medium and hard LCB v6 stdin tasks (26 medium first, then 14 hard);
- P’s Appendix A LCB prompts, adapted from functions to stdin programs, with the dataset’s hidden tests as ground truth;
- the arms ZS, SR, RO, AR (inner cap 5, first-pass termination) and ARZ, which share their prefix on each task, with an outer cap of 3 edits for the iterating arms.

Its analysis script (`emmy/ar_ablation_analyse.py`) was fixed before any results came in [ledger 43, 47]. Ada’s run (`ada/stop_signal_exp.py`) only reviews and never generates code. It uses 40 HumanEvalPack Python problems, each with a canonical and a human-injected buggy solution; the runner checked that all 40 canonical solutions pass the hidden tests and none of the 40 buggy ones do. R reviews each artifact once. Each approval is followed by one critique from C and one independent review from a second reviewer R2, which serves as a non-interacting unanimity control. P’s reviewer and critic prompts are used verbatim, plus a system line asking for responses under 200 words [ledger 44, 46, 48]. Both are new experiments with a different model and different tasks, not reproductions of P [ledger 43, 50].

Neither has produced a result. On the shared single-slot server, calls took 33–131 s, and some reached the 3000-token output cap. At about 15 minutes per task, the 40 planned ablation tasks need about 10 hours. When the peer phase ended, no ablation task had finished after 16 minutes [ledger 49, 54]. Ada’s run had 6 records (3 complete artifacts) at its last status entry, and its analysis uses only complete artifacts [ledger 46, 52]. Even when complete, 40 problems can detect only large effects, so the plan is to report intervals rather than orderings [ledger 46].

The material from this round is thin in evidence. It consists of arithmetic on P’s reported aggregates, one analytic identity and two unfinished runs. It sharpens the causal question and prices part of it, but it does not answer it [ledger 54].

Objections and resolutions

Objection 1 (Tick size attributed to Q). *An enforced tick-size result had been attributed too directly to Q* [ledger 9].

Code inspection showed that Q uses a strict floating-point improvement rule for LLM orders and applies its price increment only in a check for ZI deadlock. The tick experiment therefore concerns a new message constraint, enforced and stated in the prompt, not Q as run [ledger 9, 13]. Resolved by narrowing the claim.

Objection 2 (Same constraint for ZI-C and LLMs). *ZI-C and the LLMs might not face the same effective constraint.*

Q’s exchange does not enforce reservation values, although its prompt tells LLMs never to violate them. Violations are logged, but their rate is not reported [ledger 13, 18, 20, 22]. The 0.96 result is therefore a floor for constrained strategies, conditional on LLM compliance with the prompt, not an unconditional exchange baseline. Resolved by conditioning the claim; the rate itself remains unknown.

Objection 3 (Analogy, not a cross-paper result). *The verifier twice judged the Q–P connexion to be an analogy rather than a supported cross-paper result, and requested a matched P ablation* [ledger 24, 33].

Both peers accepted the objection. In the earlier rounds they established that the supplied material lacks P’s task instances, per-task outcomes, traces, disagreement rates, orchestration and credentials, so running the ablation would have meant inventing evidence [ledger 25, 27, 29, 34, 36]. This round rechecked public availability and found nothing new [ledger 43, 44]. It started a free controlled experiment with a different model, but the experiment has not finished [ledger 49, 54]. Not resolved.

Objection 4 (Is the gap to be decomposed real?). *The causal question matters only if the gap between AR and its controls is real. Yet P’s own AR vs MARS gap lies within the noise of a single run* [ledger 45].

Accepted by both peers [ledger 46]. Any matched ablation therefore needs paired per-task outcomes, ideally with repeated seeds. This qualifies the verifier’s request rather than answering it.

Objection 5 (P’s internal numbers). *P reports MARS at 82% in Table 1 but 85% in its prose [ledger 16, 22], and it credits MARS in the prose with AR’s 43/57 hard tasks [ledger 45].*

Both mismatches were confirmed in the source. Unresolved inside P; the exact counts above use the table values.

Objection 6 (Conduct of the local runs). *Running the local jobs raised procedural problems: contention for the shared server, a verdict-parsing bug in ada’s runner, and a prompt deviation in ada’s runner.*

The jobs sent different context sizes, which made Ollama reload the model on every alternation. Ada restarted with the same `num_ctx` (16384) as emmy [ledger 48, 50]. A markdown filter logged `NEEDS_CHANGES` verdicts as invalid. The bug did not change which calls were made, because only `APPROVE` triggers C and R2, and the analysis re-parses every logged text [ledger 52]. Ada’s runner adds a brevity line and emmy’s does not, so the two runs differ in that respect. Brevity may move R’s operating point but does not change the logic of the stop-signal test [ledger 48, 50]. Resolved procedurally; none of these issues affects a reported result, since there are none yet.

Sources outside Q and P

Earlier rounds used the following:

- Gode and Sunder’s 1993 ZI-C strategy as background (<https://www.journals.uchicago.edu/doi/10.1086/261868>).
- Q’s released repository (<https://github.com/jswistak/competitive-market-simulation>), including its configurations, its notebook and its order-handling code [ledger 4, 6, 13].
- BAZAAR (<https://github.com/lechmazur/bazaar>), found by a prior-work search. It is a sealed-bid double auction that reports individual profit rather than market efficiency, so it is a different comparison and supports no novelty claim here [ledger 14].

This round added the following:

- P’s arXiv abstract page and web searches for an author repository [ledger 43, 44].
- The LiveCodeBench `code_generation_lite` release v6, at https://huggingface.co/datasets/livecodebench/code_generation_lite [ledger 43, 45].
- HumanEvalPack (<https://huggingface.co/datasets/bigcode/humanevalpack>) [ledger 44].
- The open-weight model gemma4:26b, served locally by Ollama without a paid API [ledger 43, 44].

No outside source supplied evidence about P’s own tasks or runs.

What remains open

The material is thin on the cross-paper question. The following remain open:

- No LLM intervention has finished: the two local runs are incomplete [ledger 54].
- P’s first-pass accept count and its per-task outcomes are unknown [ledger 53, 54].
- P’s code and task list are not public [ledger 43].
- P provides no uncertainty estimates [ledger 27, 45].
- Q’s LLM reservation-violation rate is unknown [ledger 18, 54].
- Two ZI-C batches by Q’s authors differ in price dispersion without explanation [ledger 14, 54].

The supported conclusion is still negative. Q supplies a standalone constrained ZI-C result. The Q–P decomposition of institution against agent (omission against commission in Q; stopping information, forced edits and edit guidance in P) is now well posed and partly quantified, but it is not a supported cross-paper result [ledger 31, 54].

Smallest settling step

The settling experiment is unchanged. On the same P tasks, compare an R-only iterate-to-clean loop, AR, and AR with a random critic whose verdicts match the observed disagreement rate. Hold outer caps and compute equal, and report paired per-task outcomes, ideally over repeated seeds. If AR beats both controls, critic information has an effect beyond re-review and stopping. Otherwise the proposed institutional decomposition is not supported [ledger 25, 27, 31, 45].

This round found two smaller steps. For P itself, the cheapest step is the first-pass accept count from P's logs, split by whether v_0 passes. A count above about 77 would show that P's stop decision carries correctness information [ledger 53]. That count would not separate R's contribution from C's. Without P's authors, the cheapest step is to let `emmy/ar_ablation.py` and `ada/stop_signal_exp.py` finish (about 10 hours on the shared local server) and apply the analysis scripts that were fixed in advance. That would settle the decomposition for gemma4:26b on LCB v6 and HumanEvalPack, within the power of 40 problems, and not for P [ledger 46, 49, 54].